Answer **all** questions in the spaces provided.

1 A toy car of mass 0.42 kg is released from rest and accelerates along a straight track towards a wall. It hits the wall and rebounds in the opposite direction. The variation with time t of the momentum p of the toy car when not in contact with the wall is shown in Fig. 1.1

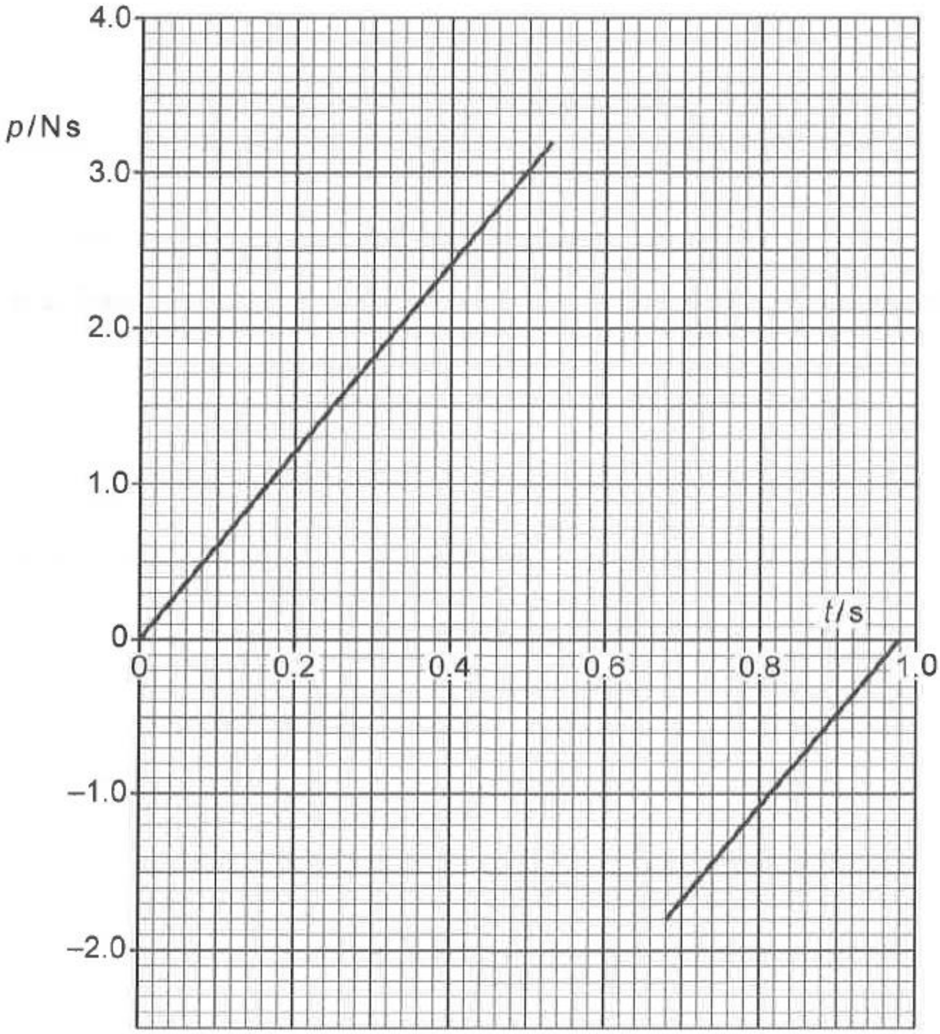

Fig 1.1

(a) Using Fig 1.1, calculate the impulse acting on the toy car during the collision.

impulse = ……………….. N s [2]

Solution:

Impulse = change in momentum

$= p_f - p_i$ M1

$= -1.80 - (3.20)$

$= -5.00$ N s A1

	(b)	Calculate the magnitude of the average acceleration of the car during the collision and state the direction of this acceleration relative to the initial motion of the car.	
		average acceleration = …………………………..m s^{-2} direction = ……………..……………………………………	[3]
		Solution: $\Delta v=\frac{\Delta p}{m}=\frac{-5.00}{0.42}=-11.9\ m\ s^{-1}$ $a=\frac{\Delta v}{\Delta t}=\frac{-11.9}{0.15}=-79.3\ m\ s^{-2}$ $\lvert a\rvert=79.3\ m\ s^{-2}$ Direction: opposite to the initial motion of the car	M1 A1 A1
	(c)	Explain why the collision was inelastic.	
		…………………………………………………………………………………………………... ………………………………………………………………………………………………….. …………………………………………………………………………………………………..	
		………………………………………………………………………………………………	[2]
		Solution: The momentum and hence speed of the car before and after the collision is different. The wall is stationary and hence has no momentum and no KE. The total initial KE and the total final KE of the system of the car and wall is not the same. Therefore, collision is inelastic.	 B1 B1 A0

	(d)	Calculate the percentage change in the kinetic energy of the car as a result of the collision.	
		percentage change = ……………………… %	[3]
		Solution: $E_{K,i}=\frac{p_i^2}{2m}=\frac{3.20^2}{2(0.42)}=12.2\ J$ and $E_{K,f}=\frac{p_f^2}{2m}=\frac{1.80^2}{2(0.42)}=3.86\ J$	M1
		$\%\Delta E_K=\frac{E_{K,f}-E_{K,i}}{E_{K,i}}\times 100\%$	C1
		$=\frac{3.86-12.2}{12.2}\times 100\%$ $=-68.4\%$ (including -ve sign)	A1

[Total: 10]